

Nirja Patel

☎ 8320612057 | ✉ nirja2501@gmail.com | 🌐 www.linkedin.com/in/nirja-patel-b12548348 | ⚡ leetcode.com/u/Nirja2501 | 📍 Ahmedabad

PROJECTS

FarmXpert

- Building a multi-agent AI decision-support system integrating IoT-based farm data to provide intelligent guidance for crop planning, farm operations, and market decisions.
- Designed a central Orchestrator Agent that coordinates 22+ specialized AI agents including soil health analysis, crop selection, irrigation planning, pest diagnostics, and market intelligence.
- Implementing a modular agent architecture inspired by MetaGPT/CrewAI, enabling agents to communicate through structured APIs and generate unified recommendations for farmers.
- Tech Stack: Python, FastAPI, LLM APIs, LangGraph, PostgreSQL, REST APIs, React.js

SmartJar

- Built an AI-driven financial planning application that automates income allocation and long-term savings strategies.
- Integrated Gemini Vision API to analyze financial inputs and generate personalized budgeting insights.
- Developed a full-stack architecture using React.js and Node.js to manage financial data and user interactions.
- Tech Stack: React.js, Node.js, Python, MongoDB, Gemini Vision API

GearGuard

- Developed a MERN-based asset management system to track equipment inventory, maintenance schedules, and technician assignments.
- Designed RESTful APIs using Node.js and Express.js for handling asset records and repair workflows.
- Implemented MongoDB database schemas to maintain equipment history and maintenance logs.
- Built responsive React.js interfaces to visualize asset status and maintenance updates.
- Tech Stack: React.js, Node.js & Express.js, MongoDB

CUDAS

- Developed an AI-powered preparation platform that generates structured learning plans and interview simulations for students.
- Built backend services using FastAPI to manage AI workflows and user progress tracking.
- Designed PostgreSQL database models using SQLAlchemy for efficient storage of learning data.
- Integrated Groq LLM APIs to dynamically generate questions and personalized feedback.
- Tech Stack: React.js, FastAPI, PostgreSQL, SQLAlchemy, JWT, SMTP, Groq LLM APIs

EXPERIENCE

Global Internships

Data Visualization Associate (Virtual Internship)

- Performed data cleaning and exploratory data analysis (EDA) on structured datasets.
- Developed data visualizations using NumPy, Pandas, and Matplotlib to present insights effectively.
- Created graphical reports and dashboards to communicate analytical findings clearly.

ACHIEVEMENTS

- **Winner – Codeversity National Level Hackathon** at IIT Gandhinagar
- **Government SSIP Grant Awardee** – Secured funding for FarmXpert under Student Startup & Innovation Policy (Gujarat)
- **Finalist – Smart India Hackathon (SIH 2025)**
- **Finalist – MumbaiHacks 2025**
- **Finalist – Odoo X Adani 2026 National Competition**

SKILLS

Languages: C, C++, Python, Java, JavaScript
Frontend: HTML, CSS, React.js
Backend: Node.js, Express.js, FastAPI
Databases: MongoDB, PostgreSQL, MySQL
AI Systems: LLM APIs (Gemini, Groq), Prompt Engineering, Agent-based Workflow Design
Data Science: NumPy, Pandas, Matplotlib, EDA, Data Visualization, Statistical Analysis
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS
Tools: Git, GitHub, JWT, OAuth

EDUCATION

Vishwakarma Government Engineering College <i>B.Tech in Information & Communication Technology</i>	2023 – 2027 <i>Ahmedabad, Gujarat</i>
– CGPA: 9.22	
Higher Secondary (12th Science Stream) <i>GSEB (English Medium)</i>	2023
– Score: 77.84% Percentile: 95.19	
Secondary School (10th) <i>GSEB (English Medium)</i>	2021
– Score: 82.17% Percentile: 92.73	